

Replication: Comparing Experimental and Nonexperimental Methods

Comparing Experimental and Nonexperimental Methods Four Decades After LaLonde (1986)

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1 Benchmark setup — NSW, LDW, CPS/PSID, and treatment timing

Research Question:

When estimating the ATT of NSW job training using nonexperimental CPS/PSID controls, do modern methods—propensity scores, trimming, matching, weighting, DML, AIPW—produce stable estimates, and do placebo tests support interpreting those estimates causally?

The empirical object is the National Supported Work (NSW) job-training evaluation. The benchmark comes from randomized assignment: eligible applicants were assigned either to a control condition or to a treatment offering up to 12 months of supported work, supervision, counseling, and placement help.

The replication note uses the LaLonde-Dehejia-Wahba (LDW) male samples as the main spine because they contain two pretreatment earnings histories. The key comparison is not “training versus no training” in the abstract; it is whether nonexperimental controls can reproduce the experimental benchmark for treated NSW participants.

Timing and variables

- $W_i = 1$: individual participated in NSW training.
- Y_i : real earnings in 1978, the post-treatment outcome.
- X_i : pretreatment covariates: age, schooling, high-school dropout/no-degree, married, Black, Hispanic, 1974 earnings, 1975 earnings, 1974 unemployment, and 1975 unemployment.
- Y_i^{pl} : 1975 earnings, used later as a placebo outcome because it predates treatment.

Samples

- Original LaLonde male experimental sample: $N_{tr} = 297$ treated and $N_{co} = 425$ controls.
- External nonexperimental controls: CPS-SSA-1 and PSID-1.
- LDW experimental sample: $N_{tr} = 185$ treated and $N_{co} = 260$ controls.

- LDW-CPS: the LDW treated units plus CPS controls.
- LDW-PSID: the LDW treated units plus PSID controls.

Table 1: Male samples used in the LaLonde and LDW comparisons.

Variable	LaLonde Experimental		LaLonde Comparison Groups		LaLonde-Dehejia-Wahba Experimental	
	LaLonde Treated	LaLonde Control	CPS Controls	PSID Controls	LDW Treated	LDW Control
Age	24.63 (6.69)	24.45 (6.59)	33.23 (11.05)	34.85 (10.44)	25.82 (7.16)	25.05 (7.06)
Years of school	10.38 (1.82)	10.19 (1.62)	12.03 (2.87)	12.12 (3.08)	10.35 (2.01)	10.09 (1.61)
Proportion high school dropouts	0.73 (0.44)	0.81 (0.39)	0.30 (0.46)	0.31 (0.46)	0.71 (0.46)	0.83 (0.37)
Proportion married	0.17 (0.37)	0.16 (0.36)	0.71 (0.45)	0.87 (0.34)	0.19 (0.39)	0.15 (0.36)
Proportion Black	0.80 (0.40)	0.80 (0.40)	0.07 (0.26)	0.25 (0.43)	0.84 (0.36)	0.83 (0.38)
Proportion Hispanic	0.09 (0.29)	0.11 (0.32)	0.07 (0.26)	0.03 (0.18)	0.06 (0.24)	0.11 (0.31)
Real earnings in 1975 (thousand)	3.07 (4.87)	3.03 (5.20)	13.65 (9.27)	19.06 (13.60)	1.53 (3.22)	1.27 (3.10)
Proportion unemployed in 1975	0.37 (0.48)	0.42 (0.49)	0.11 (0.31)	0.10 (0.30)	0.60 (0.49)	0.68 (0.47)
Real earnings in 1974 (thousand)	NA	NA	14.02 (9.57)	19.43 (13.41)	2.10 (4.89)	2.11 (5.69)
Proportion unemployed in 1974	NA	NA	0.12 (0.32)	0.09 (0.28)	0.71 (0.46)	0.75 (0.43)

2 Estimands and comparison logic — stability is not identification

Purpose: Define what is being compared when the paper asks whether modern nonexperimental methods can recover the experimental benchmark.

The target causal estimand is the average treatment effect on the treated:

$$ATT = E[Y_i(1) - Y_i(0) \mid W_i = 1].$$

In the NSW setting, this is the average effect of job training on 1978 earnings for people who actually participated. The experimental sample estimates this object directly by comparing randomized treated and control units.

The nonexperimental samples cannot observe the treated units’ untreated counterfactual. Instead, modern methods use CPS or PSID controls to estimate the adjusted statistical comparison:

$$E[Y_i \mid W_i = 1] - E\{E[Y_i \mid W_i = 0, X_i] \mid W_i = 1\}.$$

This quantity is not automatically causal. It becomes an estimate of the ATT only if the selection-on-observables design is credible: after conditioning on pretreatment covariates X_i , CPS/PSID controls must represent what NSW treated units would have earned without training.

The paper therefore makes two comparisons:

- **Estimator-stability comparison:** Do modern methods—propensity scores, trimming, matching, weighting, DML, and AIPW—produce similar adjusted estimates once overlap is improved?
- **Causal-validity comparison:** Do placebo tests support the claim that the adjusted comparison can be interpreted as the causal ATT?

These are different questions. Agreement across estimators shows that the statistical estimand is being estimated more stably. Placebo success is what supports the stronger claim that the statistical estimand equals the causal ATT.

The design bridge has two parts: unconfoundedness and overlap. Overlap is checked before estimation because without comparable controls, methods rely on extrapolation. Unconfoundedness is probed after estimation using placebo outcomes, because it cannot be verified directly.

Table 3: Data-based female regression analogue using the LCS-PSID sample.

sample	Placebo: 1975 earnings	Placebo + age/educ/race	Post: 1979 earnings	Post + age/educ/race
Experimental controls	-18 (122)	-23 (123)	821 (308)	814 (306)
PSID-1 controls	-6,649 (307)	-4,875 (279)	-4,172 (412)	-2,455 (403)

3 Historical validation problem — old nonexperimental estimators miss the randomized benchmark

Argument: LaLonde’s original exercise shows why the NSW data became a benchmark: standard regression and selection-model estimators did not reliably reproduce the randomized experimental effect when CPS/PSID controls replaced experimental controls.

Table 2: Data-based replication of LaLonde-style male regression specifications.

sample	Placebo: 1975 earnings	Placebo + age/educ/race	Post: 1978 earnings	Post + age/educ/race	DID: 1978 - 1975 earnings	DID + age	Quasi-DID: controls for 1975 earnings	Quasi-DID + age	Full controls
NSW experimental controls	39 (379)	-39 (381)	886 (488)	824 (480)	847 (582)	852 (582)	879 (486)	876 (485)	813 (486)
CPS-SSA-1 controls	-10,585 (292)	-5,487 (364)	-8,870 (408)	-4,991 (450)	1,714 (486)	361 (496)	-1,543 (451)	-1,867 (458)	-792 (480)
PSID-1 controls	-15,997 (393)	-8,523 (507)	-15,578 (508)	-8,896 (626)	420 (529)	-896 (570)	-2,380 (640)	-3,155 (660)	-1,581 (719)

Table 4: Available-variable replication of LaLonde-style treatment-selection models.

sample	exclusion_set	estimate_se	n	selection_only_used	omitted	note
NSW experimental controls	Set 1: LaLonde-style exclusions	30,460 (37,327)	722	married	smsa, emp76, afdc75, children	Two-step probit control-function estimate; missing exclusions omitted.
NSW experimental controls	Set 2: employment/children exclusions	133,103 (79,276)	722	None	emp76, children	Two-step probit control-function estimate; missing exclusions omitted.
NSW experimental controls	No excluded variables	133,103 (79,276)	722	None	None	Two-step probit control-function estimate; missing exclusions omitted.
CPS-SSA-1 controls	Set 1: LaLonde-style exclusions	-387 (1,095)	16289	married	smsa, emp76, afdc75, children	Two-step probit control-function estimate; missing exclusions omitted.
CPS-SSA-1 controls	Set 2: employment/children exclusions	-66 (1,193)	16289	None	emp76, children	Two-step probit control-function estimate; missing exclusions omitted.
CPS-SSA-1 controls	No excluded variables	-66 (1,193)	16289	None	None	Two-step probit control-function estimate; missing exclusions omitted.
PSID-1 controls	Set 1: LaLonde-style exclusions	-1,358 (1,045)	2787	married	smsa, emp76, afdc75, children	Two-step probit control-function estimate; missing exclusions omitted.
PSID-1 controls	Set 2: employment/children exclusions	1,155 (1,300)	2787	None	emp76, children	Two-step probit control-function estimate; missing exclusions omitted.
PSID-1 controls	No excluded variables	1,155 (1,300)	2787	None	None	Two-step probit control-function estimate; missing exclusions omitted.